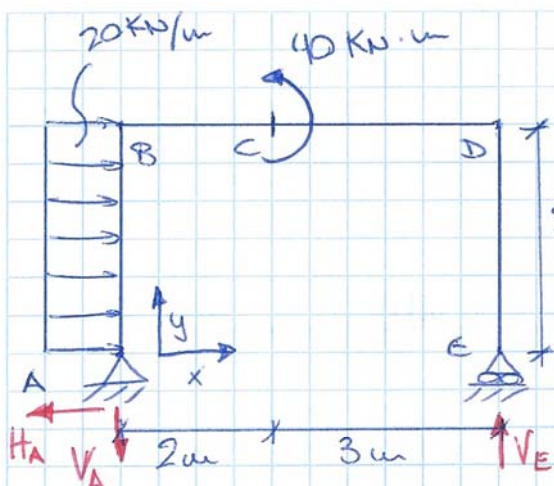


EJERCICIO 7

ESTRUCTURAS RETICULADAS



EQUILIBRIO

$$\sum F_H = 0 \Rightarrow 20 \cdot 3 - H_A = 0 \Rightarrow H_A = 60 \text{ kN}$$

$$\sum F_V = 0 \Rightarrow -V_A + V_E = 0 \Rightarrow V_A = V_E$$

$$\sum M^A = 0 \Rightarrow 5V_E + 40 - 20 \cdot 3 \cdot 1.5 = 0$$

$$V_E = \frac{50}{5} = 10 \text{ kN}$$

$$V_A = 10 \text{ kN}$$

TRAMO AB (Dorsal)

$$M(y) = H_A \cdot y - 20 \cdot y \cdot \frac{y}{2} = 60y - 10y^2 \quad \left\{ \begin{array}{l} y=0 \Rightarrow M_A = 0 \\ y=3 \Rightarrow M_B = 90 \text{ kN} \cdot \text{m} \end{array} \right.$$

$$M_{\max} = \frac{dM_y}{dy} = 0 \Rightarrow 60 - 20y = 0 \Rightarrow [y = 3]$$

$$T = -\frac{dM_y}{dy} = -60 + 20y \quad \left\{ \begin{array}{l} y=0 \Rightarrow T_A = -60 \text{ kN} \\ y=3 \Rightarrow T_B = 0 \end{array} \right.$$

$$N = V_A = 10 \text{ kN}$$

* Calculo el dintel en frontal

TRAMO BC

$$M(x) = V_E(5-x) + 40 \quad \left\{ \begin{array}{l} x=0 \Rightarrow M_B = 90 \text{ kN} \cdot \text{m} \\ x=2 \Rightarrow M_C = 70 \text{ kN} \cdot \text{m} \end{array} \right.$$

$$T = -\frac{dM(x)}{dx} = V_E = 10 \text{ kN} ; N = 0$$

Este salto de 40 kN·m
corresponde con el momento
aplicado en C

TRAMO CD

$$M(x) = V_E(5-x) \quad \left\{ \begin{array}{l} x=2 \Rightarrow M_C = 30 \text{ kN} \cdot \text{m} \\ x=5 \Rightarrow M_D = 0 \end{array} \right.$$

$$T = -\frac{dM(x)}{dx} = V_E = 10 \text{ kN} ; N = 0$$

TRAMO DE (Frontal)

$$M(y) = 0 ; T = 0 ; N = -V_E = -10 \text{ kN}$$

